

BRAC University
Department of Computer Science and Engineering

Examination: Quiz 03
Duration: 20 Minutes

Semester : Spring 25
Full Marks: 15

CSE420: Compiler Design

Name:	ID:	Section:07
-------	-----	------------

1. Draw the parse tree and evaluate the attribute values for the input string: $3 + 4 * 5n$

```
L -> {value = 0; top.push(value);}E{value = 8; top.push(value);}n{L.val = E.val}
E -> E1+ T{value = T.val;value = top.pop(); top.push(value);} {E.val = E1.val + T.val;}
E -> T {value = value + T.val;E.val = T.val;}
T -> T1 * F { top.pop(); T.val = T1.val * F.val; value = T.val + T1.val - F.val; top.push(value);}
T -> F {T.val = F.val; value = value + F.val; top.push(value);}
F -> digit {F.val = digit.lexval; value = digit.lexval * digit.lexval; top.push(value);}
```

The token for the integer numbers will be digit. Show the dependency graph for the parse tree.